NETWORK SECURITY PRACTICES – ATTACK AND DEFENSE

Reconnaissance – Vulnerability Scanning

Reconnaissance

- Systematic and methodical understanding of a system's security posture
  - Domain names, IP addresses, routers, servers,…
- Network Scanning
- **Vulnerability Scanning**
- LAN Reconnaissance
- Wireless Reconnaissance
- Custom Packet Generation

Vulnerability Scanning

- Look for known vulnerabilities in known products
- An *intrusive* test tries to exercise the vulnerability, which can crash or alter the remote target.
- A *non-intrusive* test tries not to cause any harm to the target. (by checking versions / options)

* Pros and Cons between the two?
* (intrusive) vulnerability scanning ≠ penetration test
* Good idea to use vulnerability scanning to test IDS ?

Nessus

- One of the oldest tools in vulnerability scanning (http://www.nessus.org)
- Local Vulnerabilities
  - Require login (SSH/SMB) to local machines
  - Check for security vulnerabilities, file permissions, configuration files
- Network Scan
  - Intrusive / Non-intrusive
- Client-Server architecture
  - Can control the scan engine (server) remotely
  - Scan engine can inspect multiple machines on the network

Nessus / Scan Policy

Nessus - Windows Internet Explorer

https://localhost:8834/ Certificate Error Live Search

File Edit View Favorites Tools Help

Nessus Page Tools

Nessus

am is | Help | About

Policies Reports Scans Policies Users

Add Policy

General
Credentials
Plugins
Preferences

Basic

Name: test1

Visibility: Private

Description:

Scan

- Save Knowledge Base [ ]
- Safe Checks [x]
- Silent Dependencies [x]
- Log Scan Details to Server [ ]
- Stop Host Scan on Disconnect [ ]
- Avoid Sequential Scans [ ]
- Consider Unscanned Ports as Closed [ ]
- Designate Hosts by their DNS Name [ ]

Network Congestion

- Reduce Parallel Connections on Congestion [ ]
- Use Kernel Congestion Detection (Linux Only) [ ]

Port Scanners

TCP Scan [x]	SNMP Scan [x]	Ping Host [x]
UDP Scan [ ]	Netstat SSH Scan [x]	
SYN Scan [ ]	Netstat WMI Scan [x]	

Port Scan Options

Port Scan Range: default

Performance

Max Checks Per Host	5
Max Hosts Per Scan	80
Network Receive Timeout (seconds)	5
Max Simultaneous TCP Sessions Per Host	unlimited
Max Simultaneous TCP Sessions Per Scan	unlimited

Cancel Next

Done Local intranet 100%

Nessus / Plugins

Nessus

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Policies

Reports Scans Policies Users

Edit Policy

General
Credentials
Plugins
Preferences

Filter Name

Show Only Enabled Plugins

Reset Filter

Families

- Mandriva Local Security Checks
- Misc.
- Netware
- Peer-To-Peer File Sharing
- Port scanners
- RPC
- Red Hat Local Security Checks
- SCADA
- SMTP problems

Plugins

- 28289 Ability Mail Server < 2.61 Multiple Remote DoS
- 12065 ASN.1 Multiple Integer Overflows (SMTP check)
- 11674 BaSoMail SMTP Multiple Command Remote Overflow DoS
- 19365 BusinessMail Multiple SMTP Command Remote Buffer Overflo
- 14819 Canon ImageRUNNER SMTP Arbitrary Content Printing
- 30123 Citadel SMTP makeuserkey Function RCPT TO Command Rer
- 29830 ClamAV clamav-milter black-hole-mode Sendmail Recipient Fi
- 10047 CMail MAIL FROM Command Remote Overflow
- 17985 CommuniGate Pro LISTS Module Malformed Multipart Messag

Plugin Description

overflow vulnerabilities. These issues could lead to a heap buffer overflow. A remote attacker could exploit these issues to execute arbitrary code.

This particular check sent a malformed SMTP authorization packet and determined that the remote host is not patched.

Solution

Enabled Families: 43 Enabled Plugins: 34086

Enable All Disable All

Cancel Submit

Nessus / Plugins

Nessus

amis | Help | About | Log ou

Policies

Reports Scans Policies Users

Edit Policy

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Filter Name Show Only Enabled Plugins Reset Filter

Families

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- SMTP problems

Plugins

11022	eDonkey Detection
42833	eMule IRC Module / Web Server DecodeBase16 Function Remc
11473	eMule Malformed Data Handling Remote DoS
11844	FastTrack (FT) Crafted Packet Handling Remote Overflow
35468	GigaTribe Detection
10408	Gnapster Absolute Path Name Request Arbitrary File Access
11716	Gnutella Root Directory Misconfiguration
10946	Gnutella Servent Detection
29729	iMesh P2P Client Detection

Plugin Description

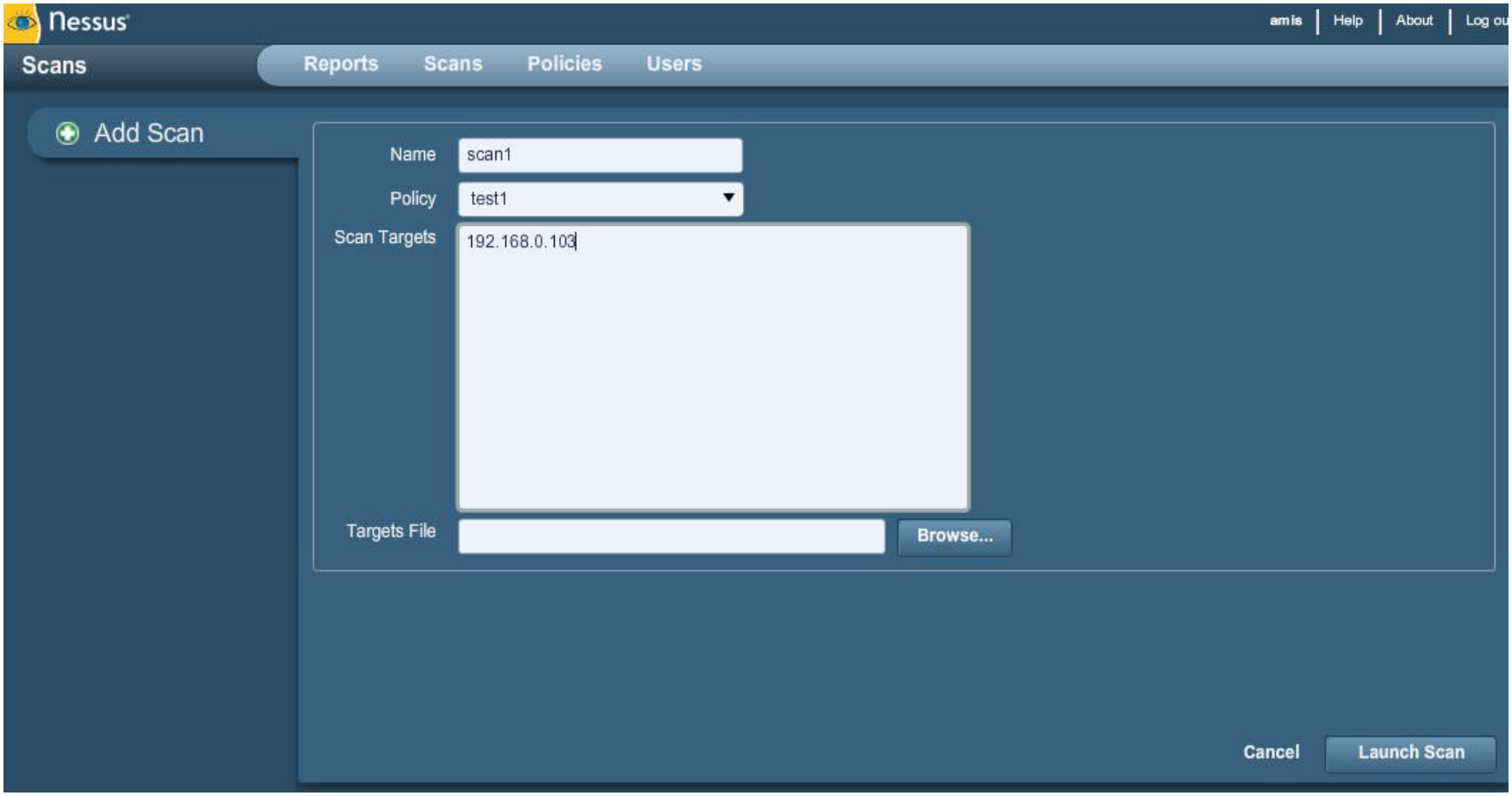
Description
According to its version, the eMule Web Server listening on this port contains a buffer overflow vulnerability in the 'DecodeBase16' function due to a lack of length checks on its inputs. An anonymous remote attacker may be able to leverage this issue to execute arbitrary code on the affected host.

Solution

Enabled Families: 43 Enabled Plugins: 34086 Enable All Disable All

Cancel Submit

Nessus / Add Scan



Nessus / Credentials

Nessus

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Policies

Reports Scans Policies Users

Edit Policy

General

Credentials

Plugins

Preferences

Credential Type: Windows credentials

SMB account : H

SMB password : ******

SMB domain (optional) :

SMB password type : Password

Additional SMB account (1) :

Additional SMB password (1) :

Additional SMB domain (optional) (1) :

Additional SMB account (2) :

Additional SMB password (2) :

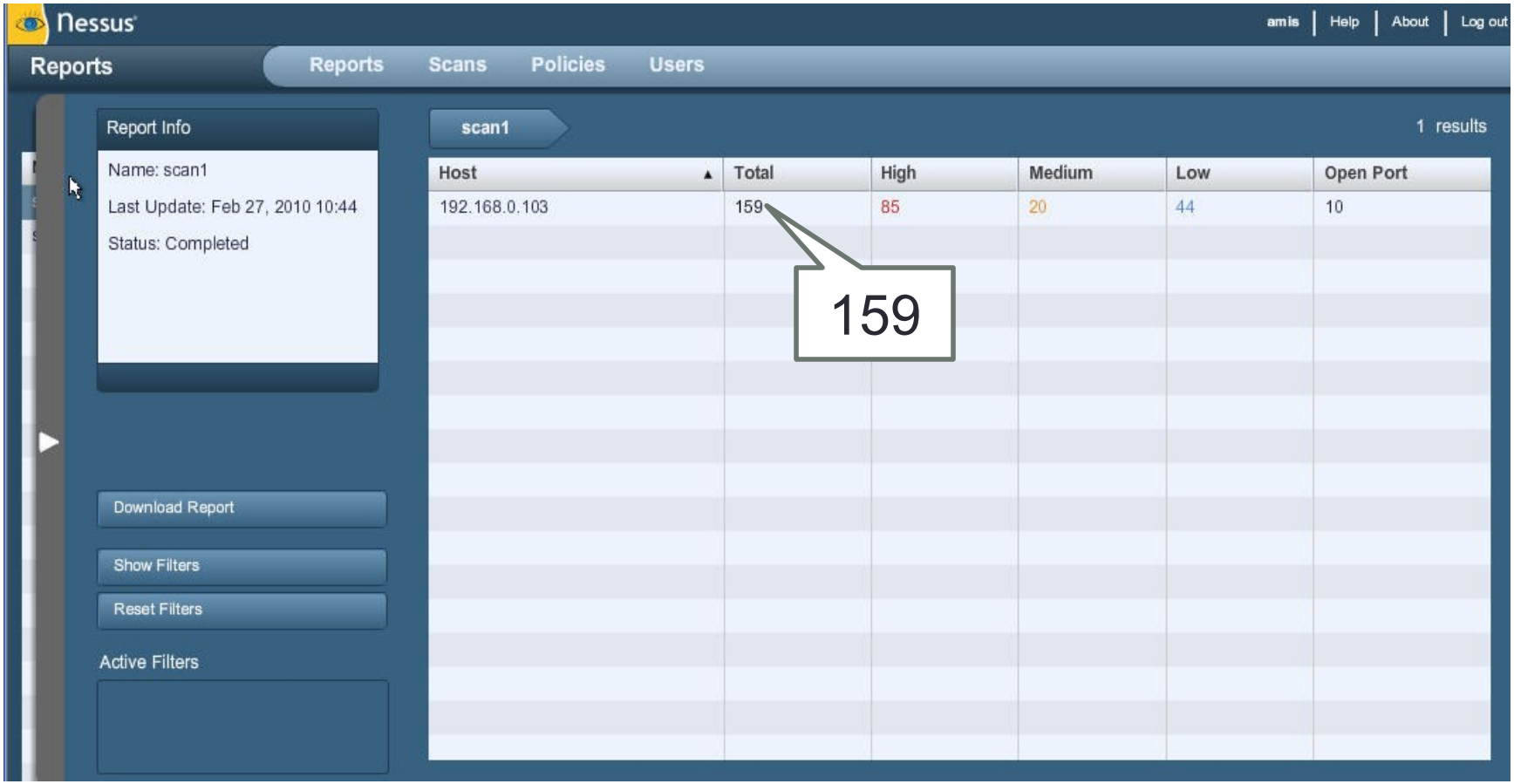
Additional SMB domain (optional) (2) :

Additional SMB account (3) :

Additional SMB password (3) :

Cancel Submit

Nessus / Scan summary



Nessus / Scan Summary (no local vulnerability check)

Nessus

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Reports

Reports Scans Policies Users

Report Info

Name: scan2 no local vulnerabil...

Last Update: Feb 27, 2010 10:57

Status: Completed

Download Report

Show Filters

Reset Filters

Active Filters

scan2 no local vulnerabilities

1 results

Host	Total	High	Medium	Low	Open Port
192.168.0.103	58	14	4	30	10

58

Nessus / Detailed scan result

MS08-067: Microsoft Windows Server Service Crafted RPC Request Handling Remote Code Execution (958644) (uncredentialed check)

Synopsis:
Arbitrary code can be executed on the remote host due to a flaw in the 'Server' service.

Description:
The remote host is vulnerable to a buffer overrun in the 'Server' service that may allow an attacker to execute arbitrary code on the remote host with the 'System' privileges.

Risk factor:
Critical

CVSS Base Score:10.0
CVSS2#AV:N/AC:L/Au:N/C:C/I:C/A:C

Solution:
Microsoft has released a set of patches for Windows 2000, XP, 2003, Vista and 2008 :
http://www.microsoft.com/technet/security/bulletin/ms08-067.mspx

Plugin ID:
34477

CVE:
CVE-2008-4250

BID:
31874

Other references:
OSVDB:49243

National Vulnerability Database

- http://nvd.nist.gov/

Sponsored by
DHS National Cyber Security Division/US-CERT

NIST
National Institute of Standards and Technology

National Vulnerability Database
automating vulnerability management, security measurement, and compliance checking

Vulnerabilities | Checklists | Product Dictionary | Impact Metrics | Data Feeds | Statistics

Home | SCAP | SCAP Validated Tools | SCAP Events | About | Contact | Vendor Comments

Mission and Overview

NVD is the U.S. government repository of standards based vulnerability management data. This data enables automation of vulnerability management, security measurement, and compliance (e.g. FISMA).

Resource Status

NVD contains:

40837 CVE Vulnerabilities
131 Checklists
191 US-CERT Alerts
2371 US-CERT Vuln Notes
2517 OVAL Queries

Last updated: 02/26/10
CVE Publication rate:
11 vulnerabilities / day

Email List

NVD provides four mailing lists to the

National Vulnerability Database Version 2.2

NVD is the U.S. government repository of standards based vulnerability management data represented using the Security Content Automation Protocol (SCAP). This data enables automation of vulnerability management, security measurement, and compliance. NVD includes databases of security checklists, security related software flaws, misconfigurations, product names, and impact metrics.

Federal Desktop Core Configuration settings (FDCC)
NVD contains content (and pointers to tools) for performing configuration checking of systems implementing the FDCC using the Security Content Automation Protocol (SCAP).
FDCC Checklists are available here (to be used with SCAP FDCC capable tools).
SCAP FDCC Capable Tools are available here.

NVD Primary Resources

- Vulnerability Search Engine (CVE software flaws and CCE misconfigurations)
- National Checklist Program (automatable security configuration guidance in XCCDF and OVAL)
- SCAP (program and protocol that NVD supports)
- SCAP Compatible Tools
- SCAP Data Feeds (CVE, CCE, CPE, CVSS, XCCDF, OVAL)
- Product Dictionary (CPE)
- Impact Metrics (CVSS)
- Common Weakness Enumeration (CWE)

NVD: Security Configuration Checklist

Home | SCAP | SCAP Validated Tools | SCAP Events | About | Contact | Vendor Comments

National Checklist Program Repository

The National Checklist Program (NCP), defined by the NIST SP 800-70 Rev. 1, is the U.S. government repository of publicly available security checklists (or benchmarks) that provide detailed low level guidance on setting the security configuration of operating systems and applications. NCP is migrating its repository of checklists to conform to the Security Content Automation Protocol (SCAP). SCAP enables standards based security tools to automatically perform configuration checking using NCP checklists. For more information relating to the NCP please visit the information page or the glossary of terms.

Search for Checklist using the fields below. The keyword search will search across the name, and summary.

Tier: Any.....
Target Product: Any.....
Product Category: Any.....
Authority: Any.....
Keyword: Search

Checklist Results

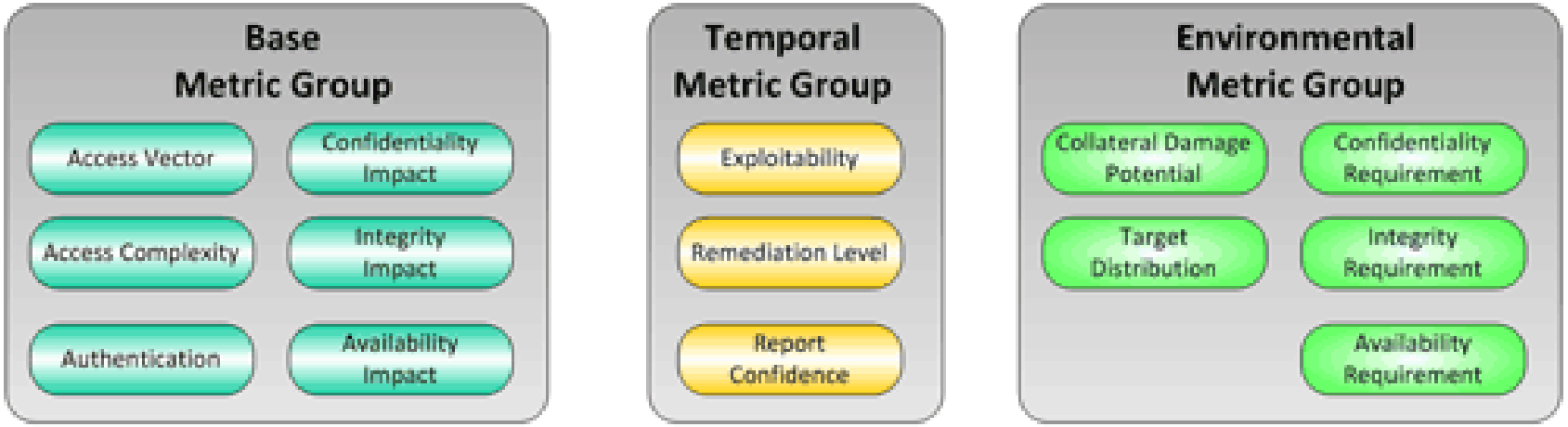
Tier	Target Product	Product Category	Authority	Publication Date	Checklist Name (Version)	Resources
IV	• Microsoft Internet Explorer 7	• Web Browser	• OMB	06/19/2008	FDCC IE7 (1.2)	• SCAP Content - OVAL 5.3 • SCAP Content - OVAL 5.4 • GPOs • Prose
IV	• Microsoft Windows Vista	• Operating System	• OMB	06/19/2008	FDCC Windows Vista (1.2)	• SCAP Content - OVAL 5.3 • SCAP Content - OVAL 5.4 • GPOs • Prose
IV	• Microsoft Windows Vista	• Operating System	• OMB	06/19/2008	FDCC Windows Vista Firewall (1.2)	• SCAP Content - OVAL 5.3 • SCAP Content - OVAL 5.4

NVD: Security Configuration Checklist

```
<definition id="oval:gov.nist.fdcc.ie7:def:1198" version="1" class="compliance">
      <metadata>
            <title>Disable Automatic Install of Internet Explorer Components</title>
            <affected family="windows">
                  <platform>Microsoft Windows XP</platform>
                  <platform>Microsoft Windows Server 2003</platform>
                  <platform>Microsoft Windows Vista</platform>
                  <product>Microsoft Internet Explorer 7</product>
            </affected>
            <reference source="http://cce.mitre.org" ref_id="CCE-3518-8"/>
            <reference source="cce.mitre.org/version/4" ref_id="CCE-684"/>
            <description>This Disable Automatic Install of Internet Explorer components setting prevents
            Internet Explorer from automatically installing components.</description>
      </metadata>
      <criteria>
            <extend_definition comment="Microsoft Internet Explorer 7 is installed"
            definition_ref="oval:gov.nist.fdcc.ie7:def:627"/>
            <criterion comment="Computer Configuration\Network\Internet Explorer\Disable Automatic Install of
            Internet Explorer Compoents" test_ref="oval:gov.nist.fdcc.ie7:tst:3956"/>
            <!--                        <criterion comment="Computer Configuration\Network\Internet
            Explorer\Disable Automatic Install of Internet Explorer Compoents"
            test_ref="oval:gov.nist.fdcc.ie7:tst:3957"/>-->
      </criteria>
</definition>
```

CVSS (Common Vulnerability Scoring System)

- http://www.first.org/cvss/cvss-guide.html
- Too many vulnerabilities to deal with
  - Prioritize the vulnerabilities and remediate the most risky ones first
- Three metric groups



CVSS / Metric Groups

- **Base**: represents the intrinsic and fundamental characteristics of a vulnerability that are constant over time and user environments.
- **Temporal**: represents the characteristics of a vulnerability that change over time but not among user environments.
- **Environmental**: represents the characteristics of a vulnerability that are relevant and unique to a particular user's environment.

CVSS Score and Vector

MS08-067: Microsoft Windows Server Service Crafted RPC Request Handling Remote Code Execution (958644) (uncredentialed check)

Synopsis:
Arbitrary code can be executed on the remote

Description:
The remote host is vulnerable to a buffer ove
execute arbitrary code on the remote host w

Risk factor:
Critical

CVSS Base Score:10.0
CVSS2#AV:N/AC:L/Au:N/C:C/I:C/A:C

Solution:
Microsoft has released a set of patches for Wi
http://www.microsoft.com/technet/security/b

Plugin ID:
34477

CVE:
CVE-2008-4250

BID:
31874

Other references:
OSVDB:49243

AV:N exploitable via *network access*
AC:L (low complexity) specialized access conditions or extenuating circumstances do not exist
Au:N no authentication required to exploit
C:C Total (complete) information disclosure
I:C Total (complete) compromise of system integrity
A:C Total (complete) shutdown of the affected resource

CVSS Score and Vector

```
BaseScore = round_to_1_decimal(((0.6*Impact)+(0.4*Exploitability)-1.5)*f(Impact))

Impact = 10.41*(1-(1-ConfImpact)*(1-IntegImpact)*(1-AvailImpact))

Exploitability = 20* AccessVector*AccessComplexity*Authentication

f(impact)= 0 if Impact=0, 1.176 otherwise

AccessVector     = case AccessVector of
                        requires local access: 0.395
                        adjacent network accessible: 0.646
                        network accessible: 1.0

AccessComplexity = case AccessComplexity of
                        high: 0.35
                        medium: 0.61
                        low: 0.71

Authentication   = case Authentication of
                        requires multiple instances of authentication: 0.45
                        requires single instance of authentication: 0.56
                        requires no authentication: 0.704
```

Nessus Plug-in Code Example

- C:\Program Files\Tenable\Nessus\nessus\plugins\zope.nasl

```
if(description)
{
   script_id(10447);  script_version ("$Revision: 1.21 $");  script_cve_id("CVE-2000-0483");
script_bugtraq_id(1354);
script_xref(name:"OSVDB", value:"347");
 script_name(english:"Zope < 2.1.7 DocumentTemplate Unauthorized DTML Entity
Modification");
[.........................]
  script_category(ACT_GATHER_INFO);
   exit(0);
}

# The script code starts here
[.........................]

port = get_http_port(default:80);
banner = get_http_banner(port:port);

if(banner)
{
    if(egrep(pattern:"^Server: .*Zope 2\.((0\..*)|(1\.[0-6]))", string:banner))
          security_hole(port);
}
```

ACT_GATHER_INFO : the script will be launched among the first. You know it will not harm the remote computer.
ACT_ATTACK : the script will attempt to gain some privileges on the remote host. It may harm the remote system (if it tests a buffer overflow for instance)
ACT_DENIAL : the script will attempt to crash the remote host
ACT_SCANNER : the script is a port scanner

Nessus Plug-in Code Example SMTP OPEN Relay

```
[...............]
send(socket: soc, data: strcat('HELO ', src_name, '\r\n'));
smtp_recv_line(socket: soc);
for (i = 0; soc && (from_l[i] || to_l[i]); i ++)
{
  mf = strcat('MAIL FROM: <', from_l[i], '>\r\n');
  send(socket: soc, data: mf);
  l = smtp_recv_line(socket: soc);
  if (! l || l =~ '^5[0-9][0-9]')
  {
    smtp_close(socket: soc);
    soc = smtp_open(port: port, helo: domain);
  }
  else
  {
    rt = strcat('RCPT TO: <', to_l[i], '>\r\n');
    send(socket: soc, data: rt);
[.................................................]
    l = smtp_recv_line(socket: soc);
    if (l =~ '^2[0-9][0-9]')
    {
      flag = 1;
[...............]
```

Smtp_relay2.nasl

HELO localhost
MAIL FROM: <nessus@localhost>
RCPT TO: <nessus@domain.com>

Check for 200~299 response code (server agrees to deliver the email)

Nessus Plug-in Code Example / Horde File Disclosure

```
  script_category(ACT_ATTACK);

[.....................................]
  # Try to exploit the issue to read a file.
  #
  # nb: Horde 3.x uses "/services"; Horde 2.x,
"/util".
  foreach subdir (make_list("/services", "/util"))
  {
    if ("util" >< subdir) file = "horde.php";
    else file = "conf.php";

    r = http_send_recv3(method:"GET",
      item:string(
        dir, subdir, "/go.php?",
        "url=../config/", file, "%00:/&",
        "untrusted=1"
      ),
      port:port
    );
[.....................................]
```

horde_url_file_disclosure.nasl

Actually exploit the vulnerability to see if *go.php* allows accessing arbitrary files (horde's config files)

Other Vulnerability Scanners

- Nikto
  - Dangerous files / CGIs, outdated versions of servers, misconfigurations in a web server
- HP WebInspect
  - Web Application Vulnerability Scanning
    - Cover Adobe Flash, JavaScript/AJAX, .Net, PHP, Cold Fusion,…

Summary

- The scanners covered thus far interact with the interfaces of a production system to detect vulnerabilities
  - Check for banners and versions
  - Look for files / settings that make system vulnerable
  - Exploit vulnerability and see if it works
- Deal with known vulnerabilities

Alternative Approaches for Vulnerability Scanning

- Static code analysis for vulnerabilities
  - http://en.wikipedia.org/wiki/List_of_tools_for_static_code_analysis
  - http://www.armorize.com/
- Software testing
  - Black-box / grey-box / white-box testing
  - http://en.wikipedia.org/wiki/Software_testing#Testing_methods
- Can identify “potential” vulnerabilities